

- Gauss (Gaus-Jordan) Yok Etme Metodu -

Verilen lineer denklem sisteminin çözümünü veya çözümlerini sistemin ilaveli matrisinin eselon forma getirilmesi yardımıyla elde edilmesine **Gauss yok etme metodu** denir. ilaveli matris eselon forma getirildikten sonra **geriye yerine koyma yöntemi** uygulanır.

Not: ilaveli matris satırca indirgenmiş eselon forma getirilirse geriye yerine koyma işleminde ciddi kolaylık sağlanır.

$$\left. \begin{array}{l} 1) \quad x - 3y = -2 \\ -x + 2y - z = -1 \\ -2y + 3z - 4t = 16 \\ -2z + 3t = -13 \end{array} \right\} \text{ lineer denklem sisteminin çözümünü bulunuz.}$$

Çözüm:

$$\left(\begin{array}{cccc|c} 1 & -3 & 0 & 0 & -2 \\ -1 & 2 & -1 & 0 & -1 \\ 0 & -2 & 3 & -4 & 16 \\ 0 & 0 & -2 & 3 & -13 \end{array} \right) \xrightarrow{\substack{R_1+R_2 \\ -R_2}} \left(\begin{array}{cccc|c} 1 & -3 & 0 & 0 & -2 \\ 0 & +1 & +1 & 0 & +3 \\ 0 & -2 & 3 & -4 & 16 \\ 0 & 0 & -2 & 3 & -13 \end{array} \right) \xrightarrow{\substack{2R_2+R_3 \\ \frac{1}{5}R_3}}$$

$$\left(\begin{array}{cccc|c} 1 & -3 & 0 & 0 & -2 \\ 0 & +1 & +1 & 0 & +3 \\ 0 & 0 & 1 & -4/5 & 22/5 \\ 0 & 0 & -2 & 3 & -13 \end{array} \right) \xrightarrow{\substack{2R_3+R_4 \\ \frac{5}{3}R_4}} \left(\begin{array}{cccc|c} 1 & -3 & 0 & 0 & -2 \\ 0 & +1 & +1 & 0 & +3 \\ 0 & 0 & 1 & -4/5 & 22/5 \\ 0 & 0 & 0 & 1 & -3 \end{array} \right) \quad \text{R.E.F.}$$

$$\left. \begin{array}{l} x - 3y = -2 \\ y + z = 3 \\ z - \frac{4t}{5} = \frac{22}{5} \\ t = -3 \end{array} \right\} \rightarrow \begin{array}{l} x = 1 \\ y = 1 \\ z = 2 \end{array}$$

$z + \frac{12}{5} = \frac{22}{5} \Rightarrow z = \frac{10}{5}$

$$\begin{array}{cccc|c}
 x & y & z & t & \\
 \begin{pmatrix} 1 & -3 & 0 & 0 & -2 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & 1 & -4/5 & 2/5 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix} & \xrightarrow{\frac{4}{5}R_4 + R_3} & \begin{pmatrix} 1 & -3 & 0 & 0 & -2 \\ 0 & 1 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}
 \end{array}$$

$$\begin{array}{cccc|c}
 \begin{pmatrix} 1 & -3 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix} & \xrightarrow{-R_3 + R_2} & \begin{pmatrix} 1 & -3 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix} & \xrightarrow{3R_2 + R_1} & \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}
 \end{array}$$

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Gözetimler!

$$\begin{array}{l}
 \rightarrow \\
 \left. \begin{array}{l} x = 1 \\ y = 1 \\ z = 2 \\ t = -3 \end{array} \right\}
 \end{array}$$

$$\left. \begin{aligned} 2) \quad x+y+w &= 3 \\ 2x+3z+t &= 1 \\ 2z-t &= -4 \\ -x+5t &= 9 \end{aligned} \right\}$$

lineer denklem sisteminin çözümünü bulma.

$$\left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 3 \\ 2 & 0 & 3 & 1 & 0 & 1 \\ 0 & 0 & 2 & -1 & 0 & -4 \\ -1 & 0 & 0 & 5 & 0 & 9 \end{array} \right) \xrightarrow[\substack{-2l_1+l_2 \\ l_1+l_3}]{\substack{-2l_1+l_2 \\ l_1+l_3}} \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 3 \\ 0 & -2 & 3 & 1 & -2 & -5 \\ 0 & 0 & 2 & -1 & 0 & -4 \\ 0 & 1 & 0 & 5 & 1 & 12 \end{array} \right) \xrightarrow{l_2 \leftrightarrow l_4}$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & 5 & 1 & 12 \\ 0 & 0 & 2 & -1 & 0 & -4 \\ 0 & -2 & 3 & 1 & -2 & -5 \end{array} \right) \xrightarrow[\substack{2l_2+l_4 \\ \frac{1}{2}l_3}]{\substack{2l_2+l_4 \\ \frac{1}{2}l_3}} \left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & 5 & 1 & 12 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & -2 \\ 0 & 0 & 3 & 11 & 0 & 19 \end{array} \right) \xrightarrow[\substack{-3l_3+l_4 \\ \frac{2}{25}l_4}]{\substack{-3l_3+l_4 \\ \frac{2}{25}l_4}}$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & 5 & 1 & 12 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & -2 \\ 0 & 0 & 0 & \frac{1}{5} & 1 & 2 \end{array} \right) \xrightarrow{\text{EF?}}$$

$$x+y+w=3$$

$$y+5t+w=12$$

$$z-\frac{t}{2}=-2$$

$$t=2$$

$$z-1=-2 \Rightarrow z=-3$$

$$w=a$$

$$x+2-a+a=3$$

$$x=1$$

$$y+10+a=12$$

$$y=2-a$$

$$(s_1, s_2, s_3, s_4, s_5) = (1, 2-a, -3, 2, a), \quad a \in \mathbb{R}$$

$$(1, 2, -3, 2, 0)$$

$$(1, 0, -3, 2, 2)$$

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çözümleri

$$C.V. = \{ (1, 2-a, -3, 2, a) \in \mathbb{R}^5 : a \in \mathbb{R} \}$$

$$= \{ (x, y, z, t, w) \mid x = 1, y = 2 - a, z = -3, t = 2, w = a, a \in \mathbb{R} \}$$